

IFN647 Week 4 Workshop Tasks

The practical tasks in this workshop focus on the TF*IDF information retrieval model and the Binary Independence Model.

The first objective is to understand how the TF*IDF technique is applied to both documents and queries, including the definition and computation of document frequency (*df*).

The second objective is to understand contingency tables and the associated variables (e.g., n_i and r_i) used in the Binary Independence Model to represent the distribution of terms in relevant and non-relevant documents. These concepts form the foundation of the BM25 ranking model.

Task 1 TF*IDF is a term-weighting scheme that combines term frequency (*tf*) and inverse document frequency (*idf*) to quantify the importance of a term within a document relative to a document collection. While the underlying intuition of TF*IDF is consistent, multiple formulations exist that differ in how term frequency, inverse document frequency, and document length normalization are defined.

weighting scheme	document term weight	query term weight
1	$f_{t,d} \cdot \log \frac{N}{n_t}$	$\left(0.5 + 0.5 \frac{f_{t,q}}{\max_t f_{t,q}}\right) \cdot \log \frac{N}{n_t}$
2	$1 + \log f_{t,d}$	$\log \left(1 + \frac{N}{n_t}\right)$
3	$(1 + \log f_{t,d}) \cdot \log \frac{N}{n_t}$	$(1 + \log f_{t,q}) \cdot \log \frac{N}{n_t}$

Discuss the TF*IDF weighting scheme recommended below and the scheme presented in the lecture notes. Your discussion should compare the two approaches with respect to:

- how *term frequency* is computed (e.g., raw counts, normalized frequency, or logarithmic scaling),
- how *inverse document frequency* is defined (e.g., smoothed vs. unsmoothed formulations),

- and how *document length normalization* is handled (e.g., cosine normalization, length-normalized TF, or no normalization).

Task 2 Manually calculate the document frequency df value for each term in the following table.

Document	term1	term2	term3	term4	term5
D1	3	0	0	5	7
D2	5	3	4	6	0
D3	0	0	5	4	6
D4	9	0	0	1	2
D5	0	1	0	3	2
D6	3	0	2	4	4
df					

Task 3 Design a python function $c_df(docs)$ to calculate df value for each term in $docs$ to verify if you can get the same result as you did in Task 2. The function returns a {term:df, ...} dictionary. In your program, you can represent the above table as follows when you use it to test your python function.

```
docs = {'D1':{'term1':3, 'term4':5, 'term5':7}, 'D2':{'term1':5, 'term2':3, 'term3':4, 'term4':6}, 'D3':{'term3':5, 'term4':4, 'term5':6}, 'D4':{'term1':9, 'term4':1, 'term5':2}, 'D5':{'term2':1, 'term4':3, 'term5':2}, 'D6':{'term1':3, 'term3':2, 'term4':4, 'term5':4}}
```

Task 4 Let $Q = \{US, ECONOM, ESPIONAG\}$ be a query, and

$C = \{D_1, D_2, D_3, D_4, D_5, D_6, D_7\}$ be a collection of documents, where

$D_1 = \{GERMAN, VW\}$
 $D_2 = \{US, US, ECONOM, SPY\}$
 $D_3 = \{US, BILL, ECONOM, ESPIONAG\}$
 $D_4 = \{US, ECONOM, ESPIONAG, BILL\}$
 $D_5 = \{GERMAN, MAN, VW, ESPIONAG\}$
 $D_6 = \{GERMAN, GERMAN, MAN, VW, SPY\}$
 $D_7 = \{US, MAN, VW\}$

Assume relevant and non-relevant documents (user feedback) are labeled as follows:

Document ID	Terms: d_{ij}	Relevance to Q
D_1	GERMAN, VW	0 no
D_2	US, US, ECONOM, SPY	1 yes
D_3	US, BILL, ECONOM, ESPIONAG	1 yes
D_4	US, ECONOM, ESPIONAG, BILL	1 yes
D_5	GERMAN, MAN, VW, ESPIONAG	0 no
D_6	GERMAN, GERMAN, MAN, VW, SPY	0 no
D_7	US, MAN, VW	0 no

For a given incoming document $D = \{US, VW, ESPIONAG\}$, let $term\ 1 = 'US'$, $term\ 2 = 'VW'$ and $term\ 3 = 'ESPIONAG'$. Based on binary independence model, work out the missing values for the following contingency tables, where $d_i = 1$ if term i is present in the document, and 0 otherwise.

	Relevant	Non-relevant	Total
$d_1 = 1$	$r_i = 3$	$n_i - r_i = 1$	$n_i = 4$
$d_1 = 0$	$R - r_i = 0$	$(N - R) - (n_i - r_i) = N - n_i - R + r_i = 3$	$N - n_i = 3$
Total	$R = 3$	$N - R = 4$	$N = 7$

	Relevant	Non-relevant	Total
$d_2 = 1$	$r_i =$	$n_i - r_i =$	$n_i =$
$d_2 = 0$	$R - r_i =$	$N - n_i - R + r_i =$	$N - n_i =$
Total	$R =$	$N - R =$	$N =$

	Relevant	Non-relevant	Total
$d_3 = 1$	$r_i =$	$n_i - r_i =$	$n_i =$
$d_3 = 0$	$R - r_i =$	$N - n_i - R + r_i =$	$N - n_i =$
Total	$R =$	$N - R =$	$N =$